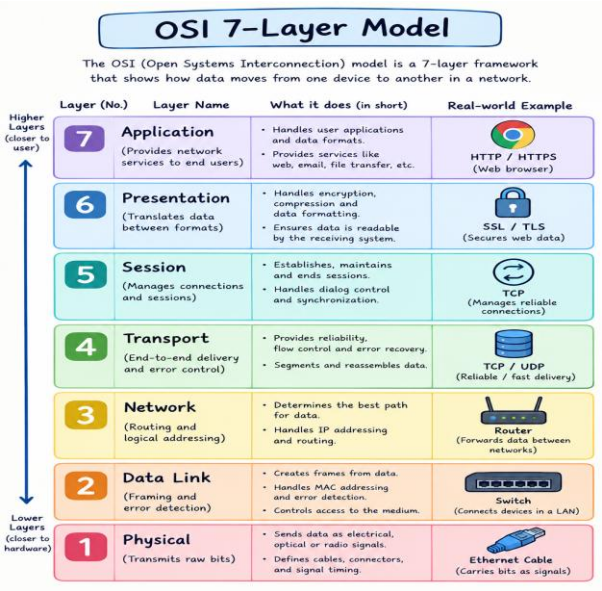
OSI LAYER

1.Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

ANS:



**2.Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.**

ANS:

a) Encrypting data before sending: Presentation (Layer 6)

b) Routing packets across networks: Network (Layer 3)

c) Converting data to electrical signals: Physical (Layer 1)

d) Opening a web page in your browser: Application (Layer 7)

e) Detecting and correcting errors in frames: Data Link (Layer 2)

3.Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

ANS:

Example: WhatsApp

When I send a message on WhatsApp, different OSI layers work together:

1. Application Layer (Layer 7): WhatsApp allows me to type and send messages to another person.
2. Transport Layer (Layer 4): TCP or other transport mechanisms help transfer data between devices reliably where applicable.
3. Network Layer (Layer 3): IP addresses help route data packets across networks to the destination.
4. Data Link Layer (Layer 2): Wi-Fi or mobile network links carry data frames between nearby network devices.
5. Physical Layer (Layer 1): Radio signals or electrical signals transmit the data through the network.

Conclusion: These OSI layers work together to deliver WhatsApp messages from the sender to the receiver.

4.Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

ANS:

OSI Layers Responsible for the Problem

Possible layers:

1. Application Layer (Layer 7): The website or web server may be down, or the browser may have a problem.
2. Presentation Layer (Layer 6): An SSL/TLS certificate or encryption problem may prevent the website from opening securely.
3. Session Layer (Layer 5): A problem establishing or maintaining the web session may interrupt access.
4. Transport Layer (Layer 4): HTTP/HTTPS traffic may be blocked by a firewall, or ports 80 and 443 may be unavailable.